Final report: environment and history through tree rings

Introductory course / proposed report assignment. Check the course instructions before submission.

Assignment

Choose one question that interests you and write either a research plan or a research review related to dendrochronology. Define the region, period, materials and tree-ring indicators as appropriate. Explain the evidence, alternative explanations, and limits of the conclusion or plan.

- Use 2–4 A4 pages and cite at least three references. A figure or table discussed in the text is encouraged.
- Absence of a figure or table does not automatically disqualify a report or result in a zero; citations and sources are also assessed.
- Identify the sources of literature, data and figures so readers can find them. Check the conditions for reproducing figures and using data.
- Follow the separate course instructions for the submission method and deadline. This document does not set them.

Two formats

Research plan: state the question, materials, methods, competing hypotheses, expected observations, feasibility and limits. Do not describe proposed measurements or expectations as completed results.

Research review: state the question and literature selection; compare the materials, methods, findings and limits of the selected studies. Explain similarities and disagreements rather than listing summaries.

Both formats use the same five criteria. Collecting original data or conducting a new experiment is not required. No extra points are awarded simply for exceeding the reference minimum or obtaining papers free of charge.

Figures, sources and the assessment proposal

Figure and table examples

A review can include an original comparison table listing the region, dating target, tree-ring indicator, main finding and limitation of at least three studies. Cite the source for each row and explain similarities and differences in the text. This is an example; do not present invented values as measurements.

A plan can show the proposed path from material to crossdating, comparison evidence and interpretation in an original conceptual diagram. Label planned work and hypotheses clearly. An arrow in a hypothesis diagram does not prove causation. Give each figure or table a number, title, scope, source and units where needed.

Finding sources

Possible starting points include Hinoki and early spring temperature (2006), northeast Asian spring temperature (2013), containers and wooden slips (2024), and the Nara institute's dataset announcement. Select sources suited to your own question.

For DOI links, open copies, libraries and requests to corresponding authors, see Supplement BG06.

How to read the proposed rubric

The course grade is based solely on the report. The following is a proposed 100-point marking scheme. There are five criteria, each worth 20 points, for 100 points total. Choose the best matching level of 20, 15, 10 or 5 points for each criterion. Award 0 when there is no relevant work for that criterion.

Proposed rubric: question, sources and comparison

1 Question and scope (20 points)

20: A specific question clearly identifies the relevant region, period and material and its relationship to tree-ring research.
15: The question and scope are clear, with a minor gap in scope or relevance.
10: A question is identifiable, but its scope is broad or what will be judged is unclear.
5: A topic is present, but the question or scope is difficult to identify.

2 Materials and methods or selection of literature (20 points)

20: Materials and methods, or the literature selection, address the question and specify concrete procedures for implementation or checking.
15: The match and procedures are generally sound, with a minor gap.
10: Materials or methods are named, but reasons for selection or procedural links are weak.
5: Names of materials or methods are given, with little connection to the question.

3 Comparison and reasoning (20 points)

20: Competing explanations or multiple studies are compared concretely, with a clear path from evidence to claims.
15: Comparison and evidence are provided, with some gaps in the differences or links to conclusions.
10: Multiple explanations or studies are mentioned, but the basis for comparison or reasoning is underdeveloped.
5: Information is mostly listed, with little clear comparison or support for claims.

Proposed rubric: presentation and limits

4 Figures, citations and presentation of sources (20 points)

20: At least three identifiable references are clearly connected to evidence in the text. If a figure or table is used, its relevance, scope, necessary units and source are also clear.
15: At least three traceable references generally support the text, with minor gaps in labels or connections; the same applies to any figure or table used.
10: Evidence or citations are present, but reference count, sources or links to claims have gaps. Any figure or table used has similar gaps in scope or source.
5: Evidence or references are fragmentary and sources are difficult to identify; the same applies to any figure or table used.

5 Conclusions, limits and next steps (20 points)

20: The supported scope and uncertainty are stated, with specific further evidence or tests that could change the conclusion. Plans are distinguished from results.
15: The conclusion, main limits and a next step are stated, with some gaps in explanation.
10: Conclusion and limits are mentioned, but their connection to evidence or a next step is unclear.
5: The conclusion or limits are fragmentary, with little connection to the evidence.

Before submission

Are at least three sources relevant to the question and connected to evidence in the text? If a figure or table is used, is it explained with scope and source? Are planned work and completed results distinct? Have you considered alternative explanations and stated the limits of the conclusion?